

Exposé III B: Calculations of Local Terms

2. Reduction to a vanishing theorem

2.1

Under the hypotheses of 1.1, let s (resp. t) denote the image of z in A (resp. B), and let $i : U = X - \{x\} \hookrightarrow X$ and $j : V = Y - \{y\} \hookrightarrow Y$ be the canonical inclusions. Since a_2 and b_1 are finite, one has $a_2^{-1}(y) = s$, $b_1^{-1}(x) = t$, hence

$$a_1^{-1}(U) \subset a_2^{-1}(V), \quad b_2^{-1}(V) \subset b_1^{-1}(U),$$

and there are commutative diagrams

$$\begin{array}{ccc} U & \xleftarrow{a'_1} & a_1^{-1}(U) \xrightarrow{a'_2} V \\ & & \downarrow j \\ X & \xleftarrow{a_1} & A \xrightarrow{a_2} Y \end{array} \quad \begin{array}{ccc} U & \xleftarrow{b'_1} & b_2^{-1}(V) \xrightarrow{b'_2} V \\ & & \downarrow j \\ X & \xleftarrow{b_1} & B \xrightarrow{b_2} Y, \end{array}$$

where the squares corresponding to $U \leftarrow a_1^{-1}(U) \rightarrow A \rightarrow X$ and to $b_2^{-1}(V) \rightarrow V \rightarrow Y$ are cartesian.

Lemma 2.2. To prove 1.2, it is enough to consider the following two cases:

$$\text{a) } L_x = 0, \quad M_y = 0; \quad \text{b) } L|U = 0, \quad M|V = 0.$$

Consider the decreasing filtrations on L and M defined by the subcomplexes

$$F^n L = \begin{cases} L, & n \leq 0, \\ i_! i^* L, & n = 1, \\ 0, & n > 1, \end{cases} \quad F^n M = \begin{cases} M, & n \leq 0, \\ j_! j^* M, & n = 1, \\ 0, & n > 1. \end{cases}$$

The complex $a_{2*} a_1^* L$ (resp. $b_{1*} b_2^* M$) is then filtered by

$$F^n a_{2*} a_1^* L = a_{2*} a_1^* F^n L.$$

We have $F^n a_{2*} a_1^* L = a_{2*} a_1^* L$ for $n \leq 0$, and 0 for $n > 1$; moreover, since a_2 is finite and the square is cartesian,

$$F^1 a_{2*} a_1^* L = j_!(a'_2)_!(a'_1)^* i^* L.$$

Similarly,

$$F^n b_{1*} b_2^* M = \begin{cases} b_{1*} b_2^* M, & n \leq 0, \\ i_!(b'_1)_!(b'_2)^* j^* M, & n = 1, \\ 0, & n > 1. \end{cases}$$

We may suppose that u (resp. v) is defined by a homomorphism of complexes

$$u : a_{2*} a_1^* L \longrightarrow M \quad (\text{resp. } v : b_{1*} b_2^* M \longrightarrow L).$$

The preceding formulas show that the composite

$$F^1 a_{2*} a_1^* L \longrightarrow a_{2*} a_1^* L \xrightarrow{u} M$$

(resp. $F^1 b_{1*} b_2^* M \rightarrow b_{1*} b_2^* M \xrightarrow{v} L$) factors uniquely through $F^1 M$ (resp. $F^1 L$); in other words, u (resp. v) is a homomorphism of filtered complexes. Thus u (resp. v) underlies a filtered correspondence, in the sense of (III 4.13), which we again denote by

$$u \in \mathrm{Hom}_{DF}(a_1^* L, a_2^! M), \quad v \in \mathrm{Hom}_{DF}(b_2^* M, b_1^! L).$$

By additivity of cup-products ([3] V 3.8.7), (III 4.13.1), one has

$$\langle u_z, v_z \rangle = \langle \mathrm{gr}^0 u_z, \mathrm{gr}^0 v_z \rangle + \langle \mathrm{gr}^1 u_z, \mathrm{gr}^1 v_z \rangle,$$

and

$$\langle u, v \rangle = \langle \mathrm{gr}^0 u, \mathrm{gr}^0 v \rangle + \langle \mathrm{gr}^1 u, \mathrm{gr}^1 v \rangle.$$

The lemma follows, since $\mathrm{gr}^1 L = i_! i^* L$ (resp. $\mathrm{gr}^1 M = j_! j^* M$) has zero fibre at x (resp. y), whereas $\mathrm{gr}^0 L$ (resp. $\mathrm{gr}^0 M$) is concentrated at x (resp. y).

Lemma 2.3. Under the hypotheses of 1.1, suppose that $L|U = 0$ and $M|V = 0$. Then

$$\langle u_z, v_z \rangle = \langle u, v \rangle.$$

The idea of the proof is that then u and v are direct images of correspondences between complexes concentrated at the closed points, and that the formula to be proved merely expresses the Lefschetz formula for the projections $x \rightarrow S$, $y \rightarrow S$. In view of 1.1.3, the data of 1.1 may be replaced by the following data: X (resp. Y) is a smooth S -curve endowed with a closed point x (resp. y), A (resp. B) is a separated S -curve, $a : A \rightarrow X \times_S Y$, $b : B \rightarrow X \times_S Y$ are finite S -morphisms such that $A \times_{X \times_S Y} B$ is reduced to a closed point z above (x, y) ,

$$u \in \mathrm{Hom}(a_1^* L, a_2^! M), \quad v \in \mathrm{Hom}(b_2^* M, b_1^! L),$$

with

$$L = h_{1*} L', \quad M = h_{2*} M', \quad L' \in \mathrm{ob} D_{\mathrm{ctf}}(\{x\}), \quad M' \in \mathrm{ob} D_{\mathrm{ctf}}(\{y\}),$$

where $h_1 : \{x\} \rightarrow X$ and $h_2 : \{y\} \rightarrow Y$ are the canonical inclusions. We must show that

$$\langle u_z, v_z \rangle = \langle u, v \rangle_z,$$

where

$$u_z \in \mathrm{Hom}(\mathrm{R}\Gamma(X, L), \mathrm{R}\Gamma(Y, M)), \quad v_z \in \mathrm{Hom}(\mathrm{R}\Gamma(Y, M), \mathrm{R}\Gamma(X, L))$$

are defined as in 1.1, and $\langle u, v \rangle_z$ is the value at z of the cup-product (III 4.2.7). We have $\mathrm{R}\Gamma(X, L) = \mathrm{R}\Gamma_c(X, L) = L_x$, $\mathrm{R}\Gamma(Y, M) = \mathrm{R}\Gamma_c(Y, M) = M_y$, and u_z (resp. v_z) is nothing other than the direct image u_* (resp. v_*) of u (resp. v) on S in the sense of (III 3.7.6), as is shown by the calculation procedure (III 3.5 b)).

On the other hand, if $h = h_1 \times_S h_2 : (x, y) \rightarrow X \times_S Y$ is the inclusion, and if $a' : A' \rightarrow (x, y)$, $b' : B' \rightarrow (x, y)$ denote the fibres of a , b at (x, y) , each reduced to a point, then there are canonical isomorphisms

$$h_* a'_* a'^! (DL_x \otimes_{\Lambda}^L M_y) \xrightarrow{\sim} a_* a'^! (DL \otimes_{\Lambda}^L M),$$

$$h_* b'_* b'^!(L_x \otimes_{\Lambda}^{\mathbf{L}} DM_y) \xrightarrow{\sim} b_* b^!(L \otimes_{\Lambda}^{\mathbf{L}} DM),$$

and the arrows

$$\begin{aligned} h_* : \mathrm{Hom}(a_1'^* L_x, a_2'^! M_y) &\rightarrow \mathrm{Hom}(a_1^* L, a_2^! M), \\ h_* : \mathrm{Hom}(b_2'^* M_y, b_1'^! L_x) &\rightarrow \mathrm{Hom}(b_2^* M, b_1^! L) \end{aligned}$$

(III 3.7.6) are isomorphisms. The correspondences u and v are therefore written uniquely as

$$u = h_* u', \quad v = h_* v',$$

with

$$u' \in \mathrm{Hom}(a_1'^* L_x, a_2'^! M_y), \quad v' \in \mathrm{Hom}(b_2'^* M_y, b_1'^! L_x).$$

By the Lefschetz formula (III 4.5.1) for $(h_1, h_2, A' \rightarrow A, B' \rightarrow B)$, one has

$$\langle u, v \rangle_z = \langle u', v' \rangle_z.$$

On the other hand, by the Lefschetz formula for $\{x\} \rightarrow S, \{y\} \rightarrow S$, one has

$$\langle u_*, v_* \rangle = \langle u', v' \rangle_z.$$

In view of the preceding remark, the conclusion follows by comparing these two equalities.

Lemma 2.4. To prove 1.2, one may suppose that a and b are closed immersions, and it is enough to prove that, if $L_x = 0$ and $M_y = 0$, the restriction map

$$H_A^0(X \times_S Y, DL \otimes_{\Lambda}^{\mathbf{L}} M) \longrightarrow H_z^0(B, b^*(DL \otimes_{\Lambda}^{\mathbf{L}} M)) \quad (2.4.1)$$

is zero.

Put $X \times_S Y = Z$. The closed subschemes $a' : A' \rightarrow Z, b' : B' \rightarrow Z$, images of a and b , satisfy the same hypotheses as a and b . Let u' (resp. v') be the correspondence supported on A' (resp. B') which is the direct image of u (resp. v) by the arrow deduced from the adjunction arrow $f_* f^! \rightarrow 1$ (resp. $g_* g^! \rightarrow 1$), where $f : A \rightarrow A'$ (resp. $g : B \rightarrow B'$) is the projection. It is immediate that $u_z = u'_z$ and $v_z = v'_z$. On the other hand, it follows easily from the definition of the cup-product 1.1.2 that

$$\langle u, v \rangle = \langle u', v' \rangle,$$

hence the first assertion.

By 2.2 and 2.3, one may suppose that $L_x = 0$ and $M_y = 0$, and the conclusion of 1.2 then becomes $\langle u, v \rangle = 0$. Let $P, Q \in \mathrm{ob} D_{\mathrm{ctf}}^b(Z)$. The cup-product

$$H_A^0(Z, P) \otimes H_B^0(Z, Q) \longrightarrow H_z^0(Z, P \otimes_{\Lambda}^{\mathbf{L}} Q)$$

decomposes as

$$H_A^0(Z, P) \otimes H_B^0(Z, Q) \xrightarrow{b^* \otimes 1} H_z^0(B, b^* P) \otimes H_B^0(Z, Q) \longrightarrow H_z^0(Z, P \otimes_{\Lambda}^{\mathbf{L}} Q),$$

where b^* is the restriction and the second arrow, a variant of the product considered in (SGA 4^{1/2} Cycle I 2.1), is derived from the evident product at the level of sections of sheaves. This assertion is easily checked, for example by means of the flasque resolutions of (SGA 4 XVII 4.2). If $a' : \{z\} \rightarrow B$,

$b' : \{z\} \rightarrow A$, $c : \{z\} \rightarrow Z$ are the inclusions, one may also invoke the standard decomposition of the Künneth arrow (III 4.2.1)

$$a'^! P \otimes_{\Lambda}^{\mathbb{L}} b'^! Q \longrightarrow c^! (P \otimes_{\Lambda}^{\mathbb{L}} Q)$$

into a base-change arrow

$$a'^! P \otimes_{\Lambda}^{\mathbb{L}} b'^! Q \longrightarrow a'^! b^* P \otimes_{\Lambda}^{\mathbb{L}} a^* b'^! Q,$$

followed by projection arrows

$$a'^! b^* P \otimes_{\Lambda}^{\mathbb{L}} a^* b'^! Q \longrightarrow a'^! (b^* P \otimes_{\Lambda}^{\mathbb{L}} b'^! Q) \longrightarrow a'^! b'^! (P \otimes_{\Lambda}^{\mathbb{L}} Q).$$

Taking $P = DL \otimes_{\Lambda}^{\mathbb{L}} M'$, $Q = L \otimes_{\Lambda}^{\mathbb{L}} DM$, and returning to the definition of $\langle u, v \rangle$ (III 4.2.5), one sees that it is enough to prove that $b^* u = 0$, whence the conclusion.

In view of 2.4, 1.2 will therefore follow from the following more precise result.

Proposition 2.5. Under the hypotheses of 1.2, suppose that a and b are closed immersions and that $L_x = 0$, $M_y = 0$. Let T be the strict localization of $X \times_S Y$ at $z = (x, y)$, and identify A , B with closed subschemes of T . Then the restriction arrows

$$b^* : \mathrm{R}\Gamma_A(T, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T) \longrightarrow \mathrm{R}\Gamma_z(B, b^*(DL \otimes_{\Lambda}^{\mathbb{L}} M)),$$

$$a^* : \mathrm{R}\Gamma_B(T, (L \otimes_{\Lambda}^{\mathbb{L}} DM)|T) \longrightarrow \mathrm{R}\Gamma_z(A, a^*(L \otimes_{\Lambda}^{\mathbb{L}} DM))$$

are zero.

2.6

The proof of 2.5 will occupy the rest of number 2 and numbers 3 and 4. By symmetry of the statement, it is enough to prove the assertion concerning b^* . We shall rewrite it in a form which will be more convenient. First note that

$$\mathrm{R}\Gamma(T, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T) = (DL \otimes_{\Lambda}^{\mathbb{L}} M)_z = (DL)_x \otimes_{\Lambda}^{\mathbb{L}} M_y = 0,$$

because $M_y = 0$, and similarly

$$\mathrm{R}\Gamma(B, b^*(DL \otimes_{\Lambda}^{\mathbb{L}} M)) = (DL \otimes_{\Lambda}^{\mathbb{L}} M)_z = 0.$$

Consequently, the relative cohomology triangles give isomorphisms

$$\mathrm{R}\Gamma(T - A, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T - A) \xrightarrow{\sim} \mathrm{R}\Gamma_A(T, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T)[1],$$

$$\mathrm{R}\Gamma(B - z, b^*(DL \otimes_{\Lambda}^{\mathbb{L}} M)|B - z) \xrightarrow{\sim} \mathrm{R}\Gamma_z(B, b^*(DL \otimes_{\Lambda}^{\mathbb{L}} M))[1],$$

under which the arrow b^* of 2.5, shifted by 1, is identified with the restriction

$$b^* : \mathrm{R}\Gamma(T - A, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|T - A) \longrightarrow \mathrm{R}\Gamma(B - z, (DL \otimes_{\Lambda}^{\mathbb{L}} M)|B - z). \quad (2.6.1)$$

Since $L_x = 0$, $M_y = 0$, one may write

$$L = i_! P, \quad M = j_! Q, \quad (2.6.2)$$

where $i : X - x \rightarrow X$, $j : Y - y \rightarrow Y$ are the inclusions, and $P \in \text{ob } D_{\text{ctf}}(X - x)$, $Q \in \text{ob } D_{\text{ctf}}(Y - y)$. By the duality isomorphism (SGA 4 XVIII 3.19.6), together with passage to the limit, one has

$$DL = Ri_* DP = Ri_* P,$$

hence

$$DL \otimes_{\Lambda}^{\mathbb{L}} M = Ri_* P \otimes_{\Lambda}^{\mathbb{L}} j_! Q.$$

Put $X \times_S Y = Z$, and identify $X \times \{y\}$ and $\{x\} \times Y$ with closed subschemes of Z (resp. T), which we shall simply denote by X and Y . Let

$$f' : Z - Y \hookrightarrow Z, \quad g' : Z - (X \cup Y) \hookrightarrow Z - Y,$$

$$f : T - Z \hookrightarrow T, \quad g : T - (X \cup Y) \hookrightarrow T - Y$$

be the inclusions. The Künneth formula (III 1.6.4) gives, by passage to the limit,

$$DL \otimes_{\Lambda}^{\mathbb{L}} M = Rf'_* g'_! (P \otimes_{\Lambda}^{\mathbb{L}} Q),$$

and hence

$$(DL \otimes_{\Lambda}^{\mathbb{L}} M)|_T = Rf_* g_! (P \otimes_{\Lambda}^{\mathbb{L}} Q)|_{T - (X \cup Y)}. \quad (2.6.3)$$

With this identification, 2.6.1 becomes

$$b^* : R\Gamma(T - (Y \cup A), g_!(E)|_{T - (Y \cup A)}) \longrightarrow R\Gamma(B - z, g_!(E)|_{B - z}), \quad (2.6.4)$$

where

$$E = (P \otimes_{\Lambda}^{\mathbb{L}} Q)|_{T - (X \cup Y)}.$$

In the next two numbers, we shall show that the arrow 2.6.4 is zero: first we shall reduce to the case where the cohomology sheaves of E have tame ramification, more precisely where the inertia action occurs through an ℓ -group, for a prime number $\ell \neq p$, and then we shall treat that case directly by means of a purity theorem.

3. Reduction to the tame case

3.1

Recall that the base ring Λ is annihilated by an integer prime to p , hence is a direct product of noetherian rings Λ_i , each Λ_i being annihilated by a power of a prime number $\ell_i \neq p$. For every scheme S' over S , the derived category $D(S', \Lambda)$ is the product of the categories $D(S', \Lambda_i)$, so that, in order to prove the vanishing of 2.6.4, we may suppose that Λ is annihilated by a power of a prime number $\ell \neq p$. We may also suppose that P (resp. Q) is bounded and has locally free components of finite type.

3.2

Put $X - x = U$, $Y - y = V$. Let U''/U be a finite Galois covering with group G which trivializes the components of P , let H be an ℓ -Sylow subgroup of G , let $U' = U''/H$ be the quotient covering, and let X' be the normalized strictly local trait of X in U' . Then X'/X is a finite connected covering of degree d_1 prime to ℓ , and the representation of $\pi_1(U')$ defined by the components of $P' = P|U'$ factors through a finite ℓ -group. Proceeding similarly with Q , one finds a finite connected covering Y'/Y of degree d_2 prime to ℓ , inducing on V an étale covering V' such that the representation of $\pi_1(V')$ defined by the components of $Q' = Q|V'$ factors through a finite ℓ -group. Let

$$d = d_1 e_1 = d_2 e_2 = \text{lcm}(d_1, d_2).$$

After adjoining an e_1 -th root (resp. an e_2 -th root) of a uniformizer of X' (resp. Y'), we may suppose that X' (resp. Y') has the same properties as above, and moreover that $d_1 = d_2 = d$.

Let x' (resp. y') be the closed point of X' (resp. Y'), T' the strict localization of $X' \times_S Y'$ at (x', y') , z' the closed point of T' , A' (resp. B') the inverse image of A (resp. B) in T' , and

$$E' = (P \overset{\text{L}}{\otimes}_{\Lambda} Q)|T' - (X' \cup Y') = (P' \overset{\text{L}}{\otimes}_{\Lambda} Q')|T' - (X' \cup Y').$$

We then have a commutative square of restriction arrows

$$\begin{array}{ccc} \text{R}\Gamma(T - (Y \cup A), g_!(E)|T - (Y \cup A)) & \xrightarrow{b^*} & \text{R}\Gamma(B - z, g_!(E)|B - z) \\ \downarrow & & \downarrow (1) \\ \text{R}\Gamma(T' - (Y' \cup A'), g'_!(E')|T' - (Y' \cup A')) & \xrightarrow{b'^*} & \text{R}\Gamma(B' - z', g'_!(E')|B' - z'), \end{array}$$

where $b' : B' \rightarrow T'$, $g' : T' - (X' \cup Y') \rightarrow T' - Y'$ are the inclusions. The covering T'/T is finite, of degree d^2 , and étale outside $X \cup Y$, hence induces an étale covering of degree d^2 from $B' - z'$ onto $B - z$. The composite of (1) and the trace morphism

$$\text{R}\Gamma(B' - z', g'_!(E')|B' - z') \longrightarrow \text{R}\Gamma(B - z, g_!(E)|B - z)$$

is therefore multiplication by d^2 , which is an isomorphism, since ℓ does not divide d . To prove that the arrow 2.6.4 is zero, it is therefore enough to prove that $b'^* = 0$. We are thus reduced to the case where the representation of $\pi_1(U)$ (resp. $\pi_1(V)$) defined by the components of P (resp. Q) factors through a finite ℓ -group, provided one ensures that A' , B' satisfy the general-position hypotheses of 1.2. This will follow from the following lemma.

Lemma 3.3. Under the hypotheses of 1.2, a and b being closed immersions, let X'/X , Y'/Y be finite coverings of strictly local S -traits, of the same degree d . Let A' (resp. B') denote the inverse image of A (resp. B) in $X' \times_S Y'$. Then the pairs (A', B') , (A', X') , (Y', B') are each in general position in the sense of 1.2.

The proof of 3.3 will be given after a few preliminaries.

Lemma 3.4. Let $X \xleftarrow{s_1} W \xrightarrow{s_2} Y$ be finite morphisms of S -traits, let $s = (s_1, s_2) : W \rightarrow X \times_S Y$, and let u, v, w be uniformizers of X, Y, W respectively, so that

$$s_1^* u = a w^m \pmod{w^{m+1}}, \quad s_2^* v = b w^n \pmod{w^{n+1}},$$

with $a, b \in k^*$. Then: (i) if $m < n$, $s(W)$ is tangent to X at $z = (x, y)$; (ii) if $m = n$, $s(W)$ is tangent to the subscheme of $X \times_S Y$ with equation

$$ap_2^*(v) - bp_1^*(u) = 0.$$

Let $Z = X \times_S Y$, and let W' be the reduced closed subscheme which is the image of s . In the tangent space to Z at z , the slope, with respect to the tangent to X , of the tangent to W' is the value at the origin of the function induced on W' by $p_2^*(v)/p_1^*(u)$; this is also the value at the origin of the function induced by $p_2^*(v)/p_1^*(u)$ on W , and is therefore 0 (resp. b/a) in case (i) (resp. (ii)).

In the situation of 3.4, we shall define the slope of W with respect to $((X, u), (Y, v))$ as the element of $k \cup \{\infty\}$ equal to 0 if $m < n$, to ∞ if $m > n$, and to b/a if $m = n$. This element is independent of the choice of w .

Lemma 3.5. Let $f : X' \rightarrow X$, $g : Y' \rightarrow Y$ be finite morphisms of S -traits with the same ramification index d , and let u, v, u', v' be uniformizers of X, Y, X', Y' respectively. Then there exists $C \in k^*$ such that, for every commutative diagram of finite morphisms of S -traits

$$\begin{array}{ccccc} X' & \xleftarrow{s'_1} & W' & \xrightarrow{s'_2} & Y' \\ & & \downarrow h & & \downarrow g \\ X & \xleftarrow{s_1} & W & \xrightarrow{s_2} & Y, \end{array}$$

one has $\lambda = C\lambda'^d$, where λ (resp. λ') is the slope of W (resp. W') with respect to $((X, u), (Y, v))$ (resp. $((X', u'), (Y', v'))$).

By hypothesis,

$$f^*u = \alpha u'^d \pmod{u'^{d+1}}, \quad g^*v = \beta v'^d \pmod{v'^{d+1}},$$

with $\alpha, \beta \in k^*$. We show that $C = \beta/\alpha$ has the required properties. Let w (resp. w') be a uniformizer of W (resp. W'). We have

$$\begin{aligned} s_1^*u &= aw^m \pmod{w^{m+1}}, & s_2^*v &= bw^n \pmod{w^{n+1}}, \\ s_1'^*u' &= a'w'^{m'} \pmod{w'^{m'+1}}, & s_2'^*v' &= b'w'^{n'} \pmod{w'^{n'+1}}, \\ h^*w &= \gamma w'^r \pmod{w'^{r+1}}, \end{aligned}$$

with $a, b, a', b', \gamma \in k^*$. Commutativity of the squares in the diagram implies

$$m'd = rm, \quad n'd = rn,$$

hence $m'/n' = m/n$. Suppose first that $m = n$. Then $\lambda = b/a$, $\lambda' = b'/a'$. If $N = m'd = rm = n'd = rn$, the preceding congruences give

$$(fs_1')^*u = \alpha a'^d w'^N \pmod{w'^{N+1}}, \quad (s_1h)^*u = a\gamma^m w'^N \pmod{w'^{N+1}},$$

whence $\alpha a'^d = a\gamma^m$. Similarly $\beta b'^d = b\gamma^n$, whence $\lambda = C\lambda'^d$, for $\lambda \neq 0, \infty$. But, since $m'/n' = m/n$, $\lambda = 0$ (resp. ∞) is equivalent to $\lambda' = 0$ (resp. ∞), which completes the proof.

We now prove 3.3. Let $Z = X \times_S Y$, $Z' = X' \times_S Y'$, let $\tilde{A}$ (resp. $\tilde{B}$) be the normalization of A (resp. B), and let $\tilde{A}'$ (resp. $\tilde{B}'$) be the normalization of A' (resp. B'). Since $\tilde{A}'$ (resp. $\tilde{B}'$) is also the normalization of $\tilde{A} \times_Z Z'$ (resp. $\tilde{B} \times_Z Z'$), we have commutative squares

$$\begin{array}{ccccc} \tilde{A}' & \longrightarrow & Z' & \longleftarrow & \tilde{B}' \\ \downarrow & & & & \downarrow \\ \tilde{A} & \longrightarrow & Z & \longleftarrow & \tilde{B}. \end{array}$$

Let $(x_i)_{1 \leq i \leq m}$ (resp. $(y_j)_{1 \leq j \leq n}$) be the points of $\tilde{A}$ (resp. $\tilde{B}$) above x (resp. y), and let $(x'_i)_{1 \leq i \leq m'}$ (resp. $(y'_j)_{1 \leq j \leq n'}$) be the points of $\tilde{A}'$ (resp. $\tilde{B}'$) above x' (resp. y'). Since A and B are strictly local, we have decompositions

$$\tilde{A} = \coprod_{1 \leq i \leq m} A_i, \quad \tilde{B} = \coprod_{1 \leq j \leq n} B_j, \quad \tilde{A}' = \coprod_{1 \leq i \leq m'} A'_i, \quad \tilde{B}' = \coprod_{1 \leq j \leq n'} B'_j,$$

where A_i, B_j, A'_i, B'_j are strictly local traits with respective closed points x_i, y_j, x'_i, y'_j . Choose uniformizers u, v, u', v' of X, Y, X', Y' respectively; let λ_i (resp. μ_j) be the slope of A_i (resp. B_j) with respect to $((X, u), (Y, v))$, and let λ'_i (resp. μ'_j) be that of A'_i (resp. B'_j) with respect to $((X', u'), (Y', v'))$. The general-position hypotheses on A and B are expressed by

$$\lambda_i \neq 0 \text{ for all } i, \quad \mu_j \neq \infty \text{ for all } j, \quad \lambda_i \neq \mu_j \text{ for all } i, j, \quad (1)$$

and we have to show that

$$\lambda'_i \neq 0 \text{ for all } i, \quad \mu'_j \neq \infty \text{ for all } j, \quad \lambda'_i \neq \mu'_j \text{ for all } i, j. \quad (2)$$

Let $i' \in [1, m']$, $j' \in [1, n']$, and let x_i (resp. y_j) be the image of $x'_{i'}$ (resp. $y'_{j'}$). The preceding diagram provides commutative squares of finite morphisms of S -traits

$$\begin{array}{ccc} X' \longleftarrow A'_{i'} & \longrightarrow & Y' \\ \downarrow & & \downarrow \\ X \longleftarrow A_i & \longrightarrow & Y \end{array} \quad \begin{array}{ccc} X' \longleftarrow B'_{j'} & \longrightarrow & Y' \\ \downarrow & & \downarrow \\ X \longleftarrow B_j & \longrightarrow & Y. \end{array}$$

By 3.5, there exists $C \in k^*$ such that

$$\lambda_i = C \lambda'_{i'}{}^d, \quad \mu_j = C \mu'_{j'}{}^d.$$

Thus (1) implies (2), which completes the proof of 3.3.

4. End of the proof of 1.2

4.1

To complete the proof of 2.5, hence of 1.2, it remains to establish the vanishing of 2.6.4 under the following additional hypotheses: Λ is annihilated by a power of a prime number $\ell \neq p$, and

P and Q are bounded, with locally free components of finite type, such that the corresponding representations of the fundamental groups factor through finite ℓ -groups. We have

$$g_!(E)|T - (Y \cup A) = g_{1!}(E_1),$$

where $g_1 : T - (X \cup Y \cup A) \hookrightarrow T - (Y \cup A)$ is the inclusion, and

$$E_1 = (P \otimes_{\Lambda}^L Q)|T - (X \cup Y \cup A).$$

Now E_1 is a complex with bounded, constructible, locally constant cohomology such that the corresponding representation of $\pi_1(T - (X \cup Y \cup A))$ factors through an ℓ -group. The assertion to be proved will therefore follow from the following more general statement.

Proposition 4.2. Let T be the strict localization at a closed point of a smooth S -scheme of dimension 2, let t be the closed point of T , and let A, B, X be closed subschemes of dimension 1 of T . Let $i : T - (A \cup X) \hookrightarrow T - A$ be the inclusion. Suppose that X is regular, and that A is in general position with respect to B and X ; that is, the intersection of the tangent cones at t to A and $B \cup X$ is reduced to t . Let $E \in \text{ob } D^b(T - (A \cup X))$ be such that $H^*(E)$ is constructible and locally constant, and that the corresponding representation of $\pi_1(T - (A \cup X))$ factors through an ℓ -group. Then the restriction arrow

$$R\Gamma(T - A, i_!(E)) \longrightarrow R\Gamma(B - t, i_!(E)|B - t)$$

is zero.

The proof will rely on the following lemma.

Lemma 4.3. Let Z be a connected regular noetherian scheme of dimension 2, with residual characteristics prime to ℓ , and let X, Y be regular divisors meeting transversely at a point z , giving a diagram of inclusions

$$\begin{array}{ccc} Z - (X \cup Y) & \xrightarrow{i'} & Z - Y \\ j \downarrow & & \downarrow j \\ Z - X & \xrightarrow{i} & Z \\ m \downarrow & & \downarrow m \\ Y - z & \xrightarrow{i_1} & Y. \end{array}$$

Assume the purity hypothesis

$$(P) \quad R^q j_*(\mathbb{Z}/\ell) = 0 \quad \text{for } q > 1.$$

Let $L \in \text{ob } D^b(Z - (X \cup Y), \Lambda)$ have constructible locally constant cohomology, with the corresponding representation of $\pi_1(Z - (X \cup Y))$ factoring through an ℓ -group. Then:

a) There is a canonical isomorphism

$$i_{1!} m'^* Rj_* i'_! L \xrightarrow{\sim} m^* Rj_*(i'_! L),$$

and the complex $m'^* Rj_* i'_! L$ has constructible, locally constant cohomology.

- b) Suppose that Y is connected, and let $\bar{s}$ be a geometric point of Y localized at a closed point $s \neq z$. Then the restriction arrow

$$R\Gamma(Y, m^* Rj_*(i'_! L)) \longrightarrow (Rj_*(i'_! L))_{\bar{s}} \quad (4.3.1)$$

is zero.

- c) Under the hypotheses of b), let C be a closed subscheme of dimension 1 of Z such that $Y \cap C = s$, and let $\tilde{C}$ be the strict localization of C at $\bar{s}$. Then the restriction arrow

$$R\Gamma(Z - Y, i'_! L) \longrightarrow R\Gamma(\tilde{C} - \bar{s}, i'_!(L)|_{\tilde{C} - \bar{s}}) \quad (4.3.2)$$

is zero.

The trivial isomorphism

$$i^* Rj_* i'_! L \xrightarrow{\sim} Rj_*(i'^* i'_! L) = Rj_* L$$

gives an isomorphism

$$m'^* Rj_* i'_! L \xrightarrow{\sim} i^* m^* Rj_* i'_! L,$$

hence, by adjunction, an arrow

$$i_{1!} m'^* Rj_* i'_! L \longrightarrow m^* Rj_*(i'_! L),$$

which induces an isomorphism outside z . We show that it is an isomorphism, that is, that if $\bar{z}$ is a geometric point above z , then

$$(Rj_*(i'_! L))_{\bar{z}} = 0.$$

For this, one may replace Λ by $\mathbb{Z}/\ell^\nu$, where ℓ^ν annihilates Λ , and by dévissage and passage to the limit reduce to the case where L is a locally constant $\mathbb{Z}/\ell^\nu$ -module of finite type corresponding to a representation of π_1 through an ℓ -group. Since the only finite abelian group of ℓ -torsion which is simple under the action of an ℓ -group is $\mathbb{Z}/\ell$ with trivial action, a further dévissage reduces to the case where L is the constant sheaf $\mathbb{Z}/\ell$.

Let $\tilde{Z}$ be the strict localization of Z at $\bar{z}$, $\tilde{X} = X \times_Z \tilde{Z}$, $\tilde{Y} = Y \times_Z \tilde{Z}$, and let $i' : \tilde{Z} - (\tilde{X} \cup \tilde{Y}) \hookrightarrow \tilde{Z} - \tilde{Y}$ be the inclusion. One has

$$(Rj_*(i'_!(\mathbb{Z}/\ell)))_{\bar{z}} = R\Gamma(\tilde{Z} - \tilde{Y}, i'_!(\mathbb{Z}/\ell)),$$

so it remains to prove that the restriction arrow

$$H^n(\tilde{Z} - \tilde{Y}, \mathbb{Z}/\ell) \longrightarrow H^n(\tilde{X} - z, \mathbb{Z}/\ell)$$

is an isomorphism for all n . For $n \leq 1$ this is a consequence of Abhyankar's lemma, and for $n > 1$ both sides vanish by the purity hypothesis. The first assertion of a) is therefore proved. We prove the second. By dévissage, we may suppose L concentrated in degree 0. The hypothesis (P) implies, by dévissage, that $R^q j_* L = 0$ for $q > 2$. It remains to prove constructibility and local constancy of $m'^* R^q j_* i'_! L$ for $q \leq 1$, which is standard. For constructibility, resolving L on the right by sheaves of the form $p_* p^* L$, where $p : Z' \rightarrow Z$ is a finite covering tamely ramified along Y , reduces to the case where L is constant, of value L_0 : then

$$m'^* Rj_* i'_! L = (L_0)_{Y-z}, \quad m'^* R^1 j_* i'_! L = (L_0)_{Y-z}(-1)$$

by Abhyankar. For local constancy, one has to see that the specialization arrows are isomorphisms. One may, for this, replace Λ by $\mathbb{Z}/\ell^\nu$ as above, and then reduce by dévissage to the case where L is the constant sheaf $\mathbb{Z}/\ell$; the conclusion follows from Abhyankar's lemma.

We have therefore proved a). We prove b). Let $\bar{\eta}$ be a geometric generic point of Y . Choose specialization arrows

$$a : \bar{\eta} \rightarrow Y(\bar{z}), \quad b : \bar{\eta} \rightarrow Y(\bar{s}),$$

where $Y(\bar{z})$ (resp. $Y(\bar{s})$) is the strict localization of Y at $\bar{z}$ (resp. $\bar{s}$). Put

$$M = m^* Rj_*(i'_! L).$$

The commutative square

$$\begin{array}{ccc} Y & \longrightarrow & Y(\bar{s}) \\ \uparrow & & \uparrow b \\ Y(\bar{z}) & \xrightarrow{a} & \bar{\eta} \end{array}$$

induces a commutative square

$$\begin{array}{ccc} R\Gamma(Y, M) & \xrightarrow{4.3.1} & M_{\bar{s}} \\ \downarrow & & \downarrow b^* \\ M_{\bar{z}} & \xrightarrow{a^*} & M_{\bar{\eta}}. \end{array}$$

Since $M|_{Y-z}$ has locally constant cohomology, b^* is an isomorphism. On the other hand, since $M = i_{1!} i_1^* M$, one has $M_{\bar{z}} = 0$, hence the arrow 4.3.1 is zero. It remains to prove c). We have $\tilde{C} = C \times_Z \tilde{Z}$, where $\tilde{Z}$ denotes, as above, the strict localization of Z at $\bar{z}$. Put $\tilde{Y} = Y \times_Z \tilde{Z}$. Then there is a commutative diagram of restriction arrows

$$\begin{array}{ccc} R\Gamma(Z - Y, i'_! L) = R\Gamma(Z, Rj_*(i'_! L)) & \xrightarrow{4.3.2} & R\Gamma(\tilde{C} - \bar{s}, i'_! L)|_{\tilde{C} - \bar{s}} \\ \downarrow & & \uparrow \\ R\Gamma(Y, m^* Rj_*(i'_! L)) & \xrightarrow{4.3.1} & (Rj_*(i'_! L))_{\bar{s}} = R\Gamma(\tilde{Z} - \tilde{Y}, (i'_! L)|_{\tilde{Z} - \tilde{Y}}) \end{array}$$

The vanishing of 4.3.2 therefore follows from b), completing the proof of 4.3.

We prove 4.2. Let $f : T' \rightarrow T$ be the blowing-up of the maximal ideal of T , let $D = f^{-1}(z) (\simeq \mathbb{P}_S^1)$ be the exceptional divisor, and let A', B', X' be the respective strict transforms of A, B, X . Then T' is regular, connected, of dimension 2, X' is a regular divisor meeting D transversely at one point z' , and the general-position hypotheses on A, B, X mean that A' is disjoint from $B' \cup X'$. Let E' be the inverse image of E on $T' - (A' \cup D \cup X')$, and let

$$i' : T' - (A' \cup D \cup X') \hookrightarrow T' - (A' \cup D)$$

be the inclusion. We must prove that the restriction arrow

$$R\Gamma(T' - (A' \cup D), i'_!(E')) \longrightarrow R\Gamma(B' \cup D, i'_!(E')|_{B' - (B' \cup D)}) \quad (*)$$

is zero. But B' is a disjoint union of strictly local schemes B'_j ($1 \leq j \leq n$), of dimension 1, such that for every j , $B'_j \cap D$ is reduced to a point b_j . Applying 4.3 c) to $(Z = T' - A', X = X', Y =$

$D - (D \cap A'), L = E', C = B'_j$), the hypothesis (P) being verified by virtue of the relative purity theorem (SGA 4 XVI 3), since (Z, Y) is a filtered projective limit of smooth pairs of codimension 1 over S , with affine étale transition morphisms, one obtains that the restriction arrow

$$\mathrm{R}\Gamma(T' - (A' \cup D), i'_!(E')) \longrightarrow \mathrm{R}\Gamma(B'_j - b_j, i'_!(E')|B'_j - b_j)$$

is zero. Therefore (*) is zero, which completes the proof of 4.2, hence of 2.5 and 1.2.